



Prepared with Synoptic

Investigation source pack

Rejected take-off and runway overrun, Tarlton Springs

Date of issue

09/10/2026

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Air safety report, pilot in command

The report the captain submitted the morning after the occurrence, under the operator's just culture scheme, quoted here exactly as he wrote it.

Fictional source material for the Tarlton Springs case study. Submitted by role; no name appears in the investigation file.

Report number: ASR-2026-1184

Submitted by: Pilot in command, SB512, 18/06/2026

Submitted: 19/06/2026 02:40 UTC

Scheme: Submitted under the company's just culture reporting policy. The report may inform the investigation; it is not to be quoted into a distributed document.

Text as submitted:

SB512 Tarlton Springs to Port Carrick, 18 June. Fifth sector of the day, picked up off another crew, doors closed on schedule. Aerodrome was in darkness, no other traffic on the CTAF, runway 09 dry, easterly around six knots. Flap ten, V1 112, FO was pilot flying.

The roll was completely normal. Both engines set, eighty knot check normal, I called V1. Immediately after that the FO called kangaroos and I looked up and saw two large animals already on the sealed surface ahead of us, moving from left to right across the centreline. I judged that continuing would put us airborne with damage we could not assess, at night, over country with no usable option, and I rejected the take-off. I had control from that point. We struck at least one animal about a second later. It was a heavy impact felt through the airframe with a secondary noise on the right side. I applied maximum braking. I selected reverse and got No. 1 only; No. 2 did not respond and I later understood the propeller had been damaged in the strike. It became clear before halfway that we would not stop on the remaining runway. We left the sealed surface at low speed and stopped in the graded area beyond the end.

There were no fire indications and no abnormal indications on the panel, so I did not order an evacuation. I set the parking brake, shut both engines down, told the cabin to remain seated, and called the aerodrome reporting officer on the CTAF and asked for the aerodrome emergency plan to be activated. We disembarked normally by the forward airstair onto the ground a few minutes later. The passengers were calm. No injuries were reported to me at the time. I understand one of the cabin crew

has since reported bruising.

I want to record two things plainly, because I think they matter more than my own decision.

First, I stopped above V1. I know what the procedure says and I have taught it myself. It was not a considered departure from procedure and I would not defend it as a technique. What I will say is that everything I have ever been trained on for a stop or go decision has been an engine problem, and every engine problem I have ever been given in a simulator has arrived below V1. I have never been trained on, briefed on, or asked to think about something standing on the runway after V1. I made that decision cold, in about a second, with nothing behind me.

Second, the kangaroos at Tarlton Springs are not a surprise to anyone who goes there. I have seen them on the strip in the evening myself several times and the ground staff treat it as an ordinary fact of the place. Tarlton Springs is not on the company's list of aerodromes where a runway inspection is requested before the first departure after dark. I had no prompt from any company document to think about it, and I did not think about it. That is the part of this I would like to see fixed, and it will not be fixed by anything to do with me.

Occurrence notification

The notification raised by the duty operations controller on the evening of the occurrence: what happened, the damage, the injuries, and the actions taken straight away.

Fictional. Saltbush Air, Tarlton Springs Aerodrome, the people, the registration and every document reference in this file are invented for teaching purposes. Any resemblance to a real occurrence is coincidental.

Notification reference: SBA-OCC-2026-0417

Raised by: Duty Operations Controller, Saltbush Air

Raised at: 18/06/2026 08:14 UTC (18:14 local)

Initial risk rating: High

Investigation required: Yes (risk rating Medium or above)

The occurrence

Item	Detail
Date	18/06/2026
Time	07:30:32 UTC (17:30:32 local, UTC+10)
Location	Tarlton Springs Aerodrome (XTLS), in the remote far west, about 380 nm inland of Port Carrick
Runway	09, sealed, 1,847 m x 30 m, RESA 90 m
Operator	Saltbush Air
Flight	SB512, Tarlton Springs to Port Carrick, scheduled regular public transport
Aircraft	Twin-turboprop regional aircraft, 50 seats, VH-SBQ
Persons on board	44 passengers, 2 flight crew, 1 cabin crew
Phase of flight	Take-off, ground roll
Injuries	Nil. One cabin crew member reported minor bruising to the left forearm during the precautionary disembarkation.
Damage	Substantial. Three No. 2 propeller blades damaged beyond limits, No. 2 engine shock-load inspection required, right main landing gear doors distorted, both right main tyres deflated through the fuse plugs, lower right fuselage skin abraded.
Third party	Two kangaroos killed. Runway 09/27 edge lighting undamaged. Aerodrome closed to all operations for 4 hours 20 minutes.

Brief description

SB512 was the last scheduled sector of the day, departing on schedule. The aerodrome was in darkness. Tarlton Springs has no control tower; the crew made their own broadcasts on the common traffic advisory frequency (CTAF) and had switched on the pilot-activated runway lighting before taxi.

During the take-off roll on runway 09 the first officer saw two kangaroos moving onto the runway from the northern side, past the 1,100 m point. The captain rejected the take-off at 118 kt, six knots above the calculated V1 of 112 kt. About one second later the aircraft struck both animals; the right

main landing gear and the No. 2 propeller took the impact.

The aircraft did not stop on the sealed surface. It departed the end of runway 09 at approximately 21 kt and came to rest 62 m into the 90 m runway end safety area, upright, on the extended centreline, with the nose wheel in graded gravel.

The captain shut both engines down, the cabin crew member was instructed to hold, and passengers were disembarked by the forward airstair 4 minutes 8 seconds after the aircraft stopped. No evacuation was commanded and no slides were deployed. At the captain's request on the CTAF, the aerodrome reporting officer activated the aerodrome emergency plan, and aerodrome staff attended with a vehicle; there was no fire.

Immediate actions taken

- Aerodrome closed by the aerodrome operator pending inspection and the removal of animal remains and propeller debris.
- Aircraft quarantined at Tarlton Springs. FDR and CVR removed by the operator's engineering representative at 12:40 UTC on 18/06/2026 and couriered to the operator's flight data section.
- Both flight crew stood down from duty and drug and alcohol tested; both results negative.
- The evening Tarlton Springs service suspended by the Head of Flying Operations pending the outcome of this investigation.

Flight data summary

The flight data section's readout of the recorders, which fixes the speeds, the times and the deceleration achieved during the stop.

Fictional source material for the Tarlton Springs case study.

Source: FDR and QAR readout, VH-SBQ, 18/06/2026, sector SB512.

Prepared by: Flight Data Section, Saltbush Air.

Reference: FDA-2026-0417-R1, issued 24/06/2026.

Clock: FDR time base is UTC. No clock error found against the aerodrome's CTAF recording; the two agreed to within 1 second at the keying of the crew's lining-up broadcast.

Performance entered by the crew

Take-off weight 20,800 kg, flap 10, bleeds on, runway 09 dry, headwind component 5 kt, aerodrome elevation 412 ft. V1 112 kt, VR 116 kt, V2 121 kt. Accelerate-stop distance required 1,780 m against an ASDA of 1,847 m, a margin of 67 m. The figures entered match the loadsheet and the performance tables at Operations Manual Part B appendix 4C.

Key parameters

UTC	Event	IAS	Groundspeed	Torque 1 / 2	Brake pressure	Notes
07:30:02	Power set, roll commences	0	0	98.4% / 98.5%	0 psi	Both power levers advanced together
07:30:15	80 kt cross-check	80	77	98.4% / 98.5%	0 psi	No caution or warning active
07:30:29	V1	112	109	98.4% / 98.5%	0 psi	Recorded aural "V1" on CVR
07:30:31	First officer's sighting call	116	113	98.4% / 98.5%	0 psi	No control input yet
07:30:32	Reject initiated	118	115	98.4% / 98.5%	0 psi	Both power levers to flight idle within 0.6 s
07:30:33	Impact	114	111	12% / 9%	2,180 psi	Lateral acceleration spike 0.31 g right; vertical spike 1.48 g
07:30:34	Maximum braking established	109	106	6% / 6%	2,940 psi	Anti-skid active on all four wheels
07:30:36	Reverse selected, No. 1 only	98	95	4% / --	2,940 psi	No. 2 reverse not selected; No. 2 propeller RPM decaying
07:30:44	--	58	56	4% / --	2,940 psi	Mean deceleration 07:30:34 to 07:30:44 is 2.6 m/s ²
07:30:51	Aircraft departs the sealed surface	21	20	4% / --	2,940 psi	Longitudinal deceleration falls to 1.9 m/s ² in gravel
07:30:57	Aircraft stops	0	0	4% / --	2,940 psi	62 m beyond the runway 09 end, on the extended centreline

07:31:26	Both engines shut down	0	0	-- / --	0 psi	Fuel levers to cut-off
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What the data shows

Both engines were at take-off torque, within 0.1% of each other, from power set until the power levers were retarded at 07:30:32. No engine parameter exceedance, caution or warning is recorded before that point. Recorded fuel flow, ITT and propeller RPM on both engines are within normal scatter for the take-off roll. There is no data supporting a power loss before the reject.

The recorded wind at the aircraft during the roll gives a headwind component of 4 to 6 kt throughout, consistent with the 07:00 UTC routine observation.

No tailwind component is recorded at any point.

The reject was initiated 3 seconds after V1 and at 118 kt, six knots above it.

Power lever movement to flight idle took 0.6 seconds and maximum braking was established 2 seconds after the decision. That response time is at the fast end of the operator's own recorded fleet distribution for rejected take-offs (fleet mean 3.2 seconds from decision to maximum braking).

Reverse was selected on No. 1 only. No. 2 propeller RPM decays from 07:30:33, consistent with the blade damage found on inspection, and the No. 2 reverse interlock did not make. The scheduled accelerate-stop figures assume symmetric reverse; the achieved mean deceleration of 2.6 m/s² is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition.

Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, leaving 695 m of sealed surface.

The stopping distance actually achieved from 118 kt was 757 m.

CTAF recording and cockpit voice recorder extract

A verbatim transcript of the aerodrome's CTAF recording and the cockpit voice recorder across the taxi, the roll, the reject, and the minutes after the aircraft stopped. The aerodrome has no control tower, so the CTAF holds the crew's own broadcasts.

Fictional source material for the Tarlton Springs case study. Speakers are identified by role only.

Sources: Tarlton Springs CTAF recording (the aerodrome operator's frequency recorder in the terminal, which records every transmission on the common traffic advisory frequency and every response of the aerodrome frequency response unit), and the cockpit voice recorder, VH-SBQ.

Reference: CVR-2026-0417, transcribed 25/06/2026 by the investigator with a second investigator verifying. Times are UTC from the CVR time base; the CTAF recorder's clock agreed with it to within 1 second.

Record type: Verbatim. Words in square brackets are the transcriber's note, not speech. Three dots mark a pause; "(unintelligible)" marks audio that two listeners could not agree on.

Tarlton Springs has no control tower. Crews broadcast their own intentions on the CTAF, and the runway lighting is pilot-activated: a series of transmitter keyings on the CTAF switches it on for 30 minutes.

PIC is the pilot in command, in the left seat and the pilot monitoring for this sector. FO is the first officer, in the right seat and the pilot flying.

ARO is the aerodrome reporting officer, transmitting from a handheld radio at the terminal. Lines marked [CTAF] were transmitted on the CTAF and appear on both recordings; every other line is the CVR alone.

07:21:50 [CTAF: three transmitter keyings from SB512, each about one second. Frequency response unit: "Tarlton Springs CTAF".]

07:21:56 PIC Lights are on. That's thirty minutes on the timer.

07:23:15 PIC [CTAF] Tarlton Springs traffic, Saltbush five one two, taxiing for runway zero nine, departing for Port Carrick, Tarlton Springs.

07:28:51 FO AWIS is still zero nine zero at six.

07:29:04 PIC Line up checks to the line. I'll make the call.

07:29:10 PIC [CTAF] Tarlton Springs traffic, Saltbush five one two, lining up runway zero nine for departure to Port Carrick, Tarlton Springs.

07:29:22 PIC Right, call's made, lights are on, nobody else about. Your aeroplane, your take-off.

07:29:24 FO My aeroplane, my take-off. Line up checks?

07:29:27 PIC Line up checks complete. Transponder's on.

07:29:41 FO OK. Setting power.
07:30:02 FO Power set.
07:30:04 PIC Power set, both engines good.
07:30:15 PIC Eighty knots.
07:30:16 FO Checked.
07:30:29 PIC V one.
07:30:31 FO Roos - roos, roos, on the left, they're on the -
07:30:32 PIC Stopping. I have control.
07:30:32 FO You have control.
07:30:33 [Sound consistent with a heavy impact. Duration approximately 0.4 seconds. Airframe rumble follows.]
07:30:34 FO (unintelligible) took one on the prop.
07:30:36 PIC Reverse - number two's not coming. Max brakes, max brakes.
07:30:41 FO Speed's coming down. Sixty ... fifty ...
07:30:47 PIC We're not stopping on it.
07:30:49 FO End's coming up.
07:30:51 [Sound of surface change. Increased low-frequency vibration.]
07:30:57 [Vibration ceases.]
07:30:59 PIC OK. We're stopped. Everyone alright?
07:31:01 FO Yeah. Yeah, good.
07:31:04 PIC Right. Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating.
07:31:12 FO Agreed, nothing showing.
07:31:26 [Sound consistent with both fuel levers being moved.]
07:31:34 PIC Cabin, flight deck, this is the captain, remain seated, remain seated. We have stopped on the aerodrome, there is no fire, stand by.
07:31:48 ARO [CTAF] Saltbush five one two, Tarlton Springs reporting officer, we saw you go off the end. Are you able to advise your situation?
07:31:54 PIC [CTAF] Tarlton reporting officer, Saltbush five one two, we've rejected, we've gone off the end of zero nine, we're stopped in the gravel. Forty four passengers, three crew, no injuries reported, no fire. We're going to get everyone off by the airstair. Request you activate the emergency plan and get a vehicle to us.
07:32:10 ARO [CTAF] Saltbush five one two, copied, activating the aerodrome emergency plan, vehicle on the way. Confirm no assistance required for injuries?
07:32:19 PIC [CTAF] Negative at this stage, we'll advise.
07:32:24 ARO [CTAF] All stations Tarlton Springs, runway zero nine two seven is obstructed by a disabled aircraft, the aerodrome is closed, Tarlton Springs reporting officer.

07:32:31 PIC [To first officer] Did you see them coming, or did they just turn up?

07:32:35 FO They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.

07:32:47 PIC I never had them. I was heads-in on the engine page after V one.

07:33:02 PIC Cabin, flight deck. We're going to disembark by the forward door onto the ground. It's dark and it's gravel out there, so nice and slow, no rush, no evacuation.

07:34:40 [Sound consistent with the forward airstair operating.]

Transcriber's note. The CVR captures no discussion of wildlife at any point in the pre-departure phase, and no runway inspection is requested or offered on the CTAF. The CTAF recording for the preceding 90 minutes contains no wildlife-related transmission from the aerodrome staff or from any aircraft. There was no other traffic on the frequency during SB512's taxi and departure.

Weather, light and moon

The observations and forecast issued for the aerodrome that evening, with the civil twilight and moon data for the night.

Fictional source material for the Tarlton Springs case study.

Source: Observations from the Tarlton Springs (XTLS) automatic weather station, which also feeds the aerodrome's automatic weather information broadcast (AWIS), and the forecast, retrieved 19/06/2026 from the operator's flight planning archive for flight SB512 on 18/06/2026.

Reference: MET-ARCH-SB512-180626.

Record type: Verbatim copies of the issued observations and forecast.

Observations

METAR XTLS 180700Z AUTO 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0
METAR XTLS 180630Z AUTO 09007KT 9999 FEW040 09/M01 Q1024 RMK RF00.0/000.0
METAR XTLS 180600Z AUTO 08008KT 9999 SCT040 11/M01 Q1023 RMK RF00.0/000.0
METAR XTLS 180800Z AUTO 09005KT 9999 FEW040 07/M02 Q1024 RMK RF00.0/000.0

Forecast held by the crew

TAF XTLS 180458Z 1806/1824
09008KT 9999 FEW040
FM181000 08006KT CAVOK
RMK T 11 08 06 05 Q 1023 1024 1024 1025

Light and moon

Tarlton Springs, 18/06/2026 (local time, UTC+10)
Sunset 16:58
End of civil twilight 17:25
Moonrise 22:41 (waning crescent, 14% illuminated)

Plain reading

The weather was not a factor in the usual sense. The 07:00 UTC observation, half an hour before the occurrence, gives an easterly at 6 kt, visibility 10 km or more, few cloud at 4,000 ft, temperature 8 degrees and a QNH of 1024. On runway 09 that is a headwind component of about 6 kt and no crosswind of consequence. Nothing in the sequence shows rain: the rainfall remark is zeros in every observation, and the runway was found dry.

The forecast held by the crew matched what occurred and contained no significant weather at any point in the period.

What the weather record does establish is the light. The take-off roll began at 17:30 local, five minutes after the end of civil twilight and 32 minutes

after sunset, with the moon not yet risen. The aerodrome was in full darkness with no natural ambient light on the runway strip, and the only lighting available was the runway 09/27 edge lighting, switched on by the crew through the pilot-activated lighting, which runs it at the one intensity set at the aerodrome's lighting controller. The unlit graded strip either side of the sealed surface was, on the investigator's own site visit at the same time of day on 23/06/2026, not visible from the runway centreline.

That matters for two reasons the map has to test. It is the condition under which the animals were not seen until they were on the sealed surface, and it is the period of the day in which the aerodrome's own sighting log records most of its wildlife movement.

Operator procedures and policy extracts

Verbatim extracts from the aerodrome standard, the aerodrome manual, the operations manual, the route risk assessment and the change record, transcribed from the copies current on the day.

Fictional source material for the Tarlton Springs case study. Every

document, reference number and revision date below is invented.

Collected by: Lead investigator, 22/06/2026 to 26/06/2026.

Record type: Verbatim extracts, transcribed from the controlled copies current on 18/06/2026.

1. Aerodrome Standard AS-139, section 10.4, "Wildlife hazard management"

10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome.

10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

10.4.3 Where subsection 10.4.2 does not apply, the measures taken under 10.4.1 need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place.

Tarlton Springs is a certified aerodrome. Its heaviest scheduled aeroplane is the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. Section 10.4.2 therefore does not apply to it, and it held no wildlife hazard management plan. It is not in breach of the standard.

2. Tarlton Springs Aerodrome Manual, section 6.3, "Wildlife management"

6.3 Aerodrome staff will conduct a visual inspection of the movement area before the first scheduled arrival of the day and record the result in the daily inspection log. Wildlife observed on or adjacent to the movement area will be dispersed where it is safe to do so and recorded in the wildlife sighting log. Inspections outside these times will be conducted on the request of the aerodrome reporting officer or a pilot in command.

The first scheduled arrival is at 07:50 local. There is no scheduled inspection at any other time of day, and none after last light. Tarlton

Springs has no control tower: it operates on a common traffic advisory frequency (CTAF), on which crews broadcast their own intentions, and its runway lighting is pilot-activated from the CTAF. A pilot in command wanting an inspection asks the reporting officer, in person or by telephone during a turnaround or on the CTAF from the aircraft. Aerodrome staff duty hours end at 18:30 local; after that the aerodrome is unattended except for the reporting officer on call.

3. Saltbush Air Operations Manual Part A, section 8.14, "Wildlife hazards", revision 9, 04/2024

8.14.1 Wildlife presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk.

8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

8.14.4 A wildlife strike shall be reported on an air safety report. Where a strike occurs, the Safety Department will review the aerodrome's status against the criteria in Appendix 8A.

Appendix 8A, Wildlife-active aerodromes. An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months. The list is reviewed annually by the Safety Department.

Appendix 8A as current on 18/06/2026 lists nine aerodromes. Tarlton Springs is not among them, and has never been listed. The list was last reviewed on 03/03/2025.

The gap. The only route onto the list is a reported strike. A sighting cannot put an aerodrome on the list, and nothing in Part A asks a crew to report a sighting. Tarlton Springs has had no company strike, so it cannot be listed; because it is not listed, no inspection is requested; because no inspection is requested, nothing is looked for. The control is triggered by the outcome it exists to prevent.

4. Saltbush Air Operations Manual Part B, section 4.6, "Rejected take-off", revision 14, 09/2025

4.6.2 Below 80 kt the take-off shall be rejected for any malfunction or

condition that in the judgement of the pilot in command makes the aircraft unfit to fly.

4.6.3 Between 80 kt and V1 the take-off shall be rejected only for: fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly.

4.6.4 At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached.

Section 4.6 contains no guidance on an obstruction seen on the runway, at any speed, and no amplification of 4.6.4 for a case in which continuing is itself hazardous.

5. Route Risk Assessment RRA-041, Tarlton Springs, issue 3, 11/2022

Hazards assessed: runway length and accelerate-stop margin; single runway with no lighting redundancy; no alternate within 200 nm at night; limited engineering support; heat and density altitude in summer.

Operating restriction: daylight operations only pending review of night lighting and diversion fuel policy.

Issue 3 is the current issue. No issue 4 exists.

6. Change Management Record CMR-2025-118, "Introduction of the Tarlton Springs evening service", 05/2025

Scope of change: addition of one evening rotation, Port Carrick to Tarlton Springs to Port Carrick, six days per week, from 14/07/2025.

Assessment: schedule change only. Existing aerodrome, existing aircraft type, existing crew qualifications, existing engineering support arrangements. No new hazard identified.

Route risk assessment review required: No.

Approved: Head of Flying Operations, 22/05/2025.

The change introduced the first scheduled operation into Tarlton Springs after last light. The daylight-only restriction at RRA-041 issue 3 is not referenced anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receipt of the approved record.

Maintenance, training and aerodrome records

The investigator's description of what the aircraft, crew, training and aerodrome records show, with extracts quoted where the wording matters.

Fictional source material for the Tarlton Springs case study.

Collected by: Lead investigator, 19/06/2026 to 03/07/2026.

Record type: Investigator's description of what the records show, except where an extract is quoted.

1. Aircraft maintenance history, VH-SBQ

The aircraft was released to service on 16/06/2026 following a scheduled check. The deferred defect list carried two items on 18/06/2026: a passenger service unit light in row 14 and a cabin door seal chafe mark, neither with any bearing on this occurrence.

No defect relating to brakes, anti-skid, tyres, propellers, reverse, engine control or runway lighting interface was recorded in the 90 days before the occurrence. Brake wear indicators were within limits at the 16/06 check.

2. Post-occurrence inspection, 19/06/2026 to 21/06/2026

- No. 2 propeller: three blades damaged beyond repairable limits, leading edge deformation and one blade tip separation. Animal tissue recovered from two blade roots. Engine subject to a shock-load inspection; borescope of both engines found no pre-existing damage and no evidence of any in-flight or on-ground power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. Gear doors distorted by impact. Gear structure within limits on detailed inspection.
- Brakes: brake energy absorbed calculated at 84% of the certified maximum for the weight and speed. No brake fire, no hydraulic leak. Anti-skid system tested serviceable after the occurrence.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.
- Runway and RESA survey: two animal carcasses recovered, one at 1,188 m and one at 1,221 m from the runway 09 threshold. Continuous tyre marks from 1,160 m to the runway end. Nose wheel rut in the RESA to 62 m beyond the end, on the extended centreline.

3. Cabin emergency lighting functional check, 24/06/2026

Raised following the first officer's account. With the emergency lighting system armed and the aircraft on battery power, two of the six floor

proximity escape path segments on the left side aft of row 16 did not illuminate. The fault was traced to a failed ballast common to both segments. The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs; it does not illuminate or monitor the floor proximity segments, so the defect was not detectable by any check available to the crew or to a line engineer without a dedicated functional test. The last dedicated functional test was 11 months before the occurrence, within the maintenance programme's 24-month interval.

The occurrence disembarkation was conducted with the main cabin lights on, by the forward airstair, without urgency. Nothing about the defect affected what happened on the night.

4. Flight crew records

Pilot in command. Total 8,410 hours, 2,140 on type, in command 3 years 1 month. Last cyclic simulator 11/03/2026, all items satisfactory. Last line check 02/12/2025, satisfactory. No prior occurrence history. Duty and rest in the 7 days before the occurrence within the company's limits with no variations applied.

First officer. Total 1,655 hours, 405 on type, 14 months with the company. Last cyclic simulator 28/04/2026, all items satisfactory. Two prior sectors into Tarlton Springs, both in daylight, on 09/01/2026 and 17/02/2026.

Duty record, 18/06/2026. Sign-on 06:30 local. Five sectors: Port Carrick to Hensby, Hensby to Mardle Creek and Mardle Creek to Port Carrick as rostered, then SB511 Port Carrick to

Tarlton Springs and SB512 Tarlton Springs to Port Carrick, the evening rotation, added on the morning of 18/06/2026 when a member of the rostered crew reported unfit. The occurrence was at 17:30 local, 11 hours 00 minutes into a duty planned to finish at 19:30 local (SB512 scheduled on blocks in Port Carrick at 19:15), giving a planned duty of 13 hours 00 minutes. The maximum permitted duty for five sectors with this sign-on, under the company's approved fatigue risk management system, is 13 hours 30 minutes. A ground delay in Port Carrick during the third sector was recovered on the Tarlton Springs turnaround: SB512 was scheduled off blocks at 17:20 local, the doors closed at 17:16, and it departed on schedule.

****Roster pattern, evening Tarlton Springs rotation, 14/07/2025 to 18/06/2026.**** Extracted from the rostering system. The rotation operated on 289 days in the period. On every one of them it was rostered as the fourth

and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00, planned at between 12 hours 45 minutes and 13 hours 15 minutes against the 13 hours 30 minute maximum for five sectors from those sign-on times. On six of those days a member of the rostered crew reported unfit and the rotation was reassigned to a crew already on duty, extending that crew's day from three sectors to five; 18/06/2026 was the sixth.

5. Recurrent training syllabus TRG-SYL-07, cycle 2025 to 2026

Module 4, Rejected take-off. Objective: the crew will demonstrate a correctly executed rejected take-off for an engine failure, an engine fire and a configuration warning, each initiated below V1, and will demonstrate the decision to continue for a failure recognised at or after V1.

The module's four scripted simulator scenarios are: engine failure at 90 kt; engine fire at 100 kt; configuration warning at 85 kt; engine failure at V1 plus 5 kt, continued. Every scenario is an aircraft system event. None presents an obstruction, an animal or any external hazard on the runway, at any speed.

The company's training records show 214 crew completed module 4 in the 2025 to 2026 cycle. No crew has been presented with an external obstruction scenario in the simulator in the five cycles held on file.

6. Tarlton Springs aerodrome operator records

Wildlife sighting log, 01/01/2025 to 18/06/2026. 46 entries. 41 of the 46 record kangaroos on or adjacent to the movement area. 38 of those 41 are timed between one hour before and two hours after last light. Eleven entries record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer confirmed that no airline has ever asked for it.

Aerodrome works register. Item WR-221, raised 04/2023: "Perimeter fence, northern boundary between the 700 m and 1,300 m runway markers, four panels displaced and wire strained, animals observed passing through. Recommend replacement, estimated \$41,000." Deferred at the 2023, 2024 and 2025 budget rounds, each time with the entry "defer, unfunded, review next cycle". Item WR-263, raised 02/2026, records the same section as "deteriorating further". A site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

Daily inspection log, 18/06/2026. One inspection recorded, at 07:35 local, before the first arrival. Result: "movement area clear". No further inspection was carried out that day and none was requested.

Runway lighting photometric check, 25/06/2026. Carried out at the investigator's request. The runway lighting is pilot-activated: keyed from the CTAF, it runs for 30 minutes at the single intensity set at the aerodrome's lighting controller, which was found at stage 3, the highest of the fittings' three stages and the setting recorded at commissioning. Crews cannot vary it. Four runway 09/27 edge lights on the northern side between the 900 m and 1,200 m markers measured between 118% and 131% of the nominal intensity for stage 3. The remainder were within tolerance. The aerodrome's works register and electrical log record no work on the runway lighting or its controller between 18/06/2026 and the check. The aerodrome operator's own maintenance programme checks lamp serviceability but does not measure intensity.

Interview transcript, pilot in command

The Teams transcript of the interview with the pilot in command, with every speaker identified by role.

Lead investigator: Right, we're recording now. Thanks again for coming in. Just so it's on the record, you're happy with us recording this?

Captain: Yeah, no problem.

Lead investigator: Good. So what I'd like first is just for you to take me through the day in your own words. Start wherever you want to start and don't worry about what I might be after. I'll hold my questions to the end.

Captain: Sure. So it was a three-sector day that turned into five, because we picked up the Tarlton evening run when the other crew went sick. Sign on was oh six thirty local. Port Carrick, Hensby, Mardle Creek, back to Port Carrick, then out to Tarlton and back. The Tarlton one is the long one. We were running late from about the second sector on. There was a ground delay in Port Carrick in the middle of the day, but there's slack in the Tarlton turn and we got it back. By the time we got into Tarlton we were on time, near enough. Turnaround's thirty minutes there, so. Nothing unusual on the ground. Fuelled, forty four on, bags on, and we closed up right on time. It was dark. It's dark out there early in June. There's nothing around that aerodrome, no town lights, nothing. The FO was flying the sector. Normal brief, normal figures, flap ten, dry runway zero nine. We had a good headwind, five or six knots. Nothing about the performance was tight in my head at the time. Lights on, taxi, made our calls on the CTAF, lined up, she set power. Everything normal, both engines good, eighty knots. I called V one. And then - Then she just yelled. Roos, roos. And I looked up and there were two of them, big ones, already on the seal, coming across left to right. And I stopped. I said stopping, I have control, and I stopped. We hit at least one of them almost straight away. You felt it through the airframe, it was a proper thump, and something let go on the right. Max brakes. I went for reverse and number two wasn't there. Number one came in, number two was just dead. And I remember thinking, about halfway through, we're not stopping on this. I could see the end lights coming and I knew. We went off the end doing maybe twenty knots. It's gravel out there, it's graded, we just dug in and stopped. Nothing on the board, no fire, so I didn't evacuate. Shut the engines down, told the cabin to sit tight, got the reporting officer on the CTAF. We got them off by the airstair a few minutes later. Nobody was hurt. That's it, really.

Lead investigator: That's good, thank you. Can I take you back to the turnaround at Tarlton.

Captain: Yep.

Lead investigator: Was there any discussion at any point, with anyone, about animals on the aerodrome?

Captain: No. None. And I want to be fair here, I'm not saying anyone should have. It just never came up. Nobody said anything and I didn't ask.

Lead investigator: Is there anything in the company documentation that would have prompted you to ask, that you're aware of?

Captain: There's a wildlife section in Part A. Eight something. It's got a list of aerodromes in the back where you request an inspection before the first one after dark. Tarlton's not on it. I've looked since. I'd have bet money it was on it, and it isn't.

Lead investigator: When you say you'd have bet money - why would you have expected it to be there?

Captain: Because everyone who flies out there knows there are roos. You see them on the strip. Not on the seal, out on the grass, in the lights, most evenings. I've seen them there myself, I don't know, four or five times. Daytime you never see one. Evening you do.

Lead investigator: Have you ever reported that to the company?

Captain: No. I've mentioned it to the ground staff out there, and they say yeah, they're always around. I've never put in a report about seeing a kangaroo. You'd report a strike. Nobody writes up a sighting. I don't think I've ever seen anyone do that.

Second investigator: Can I just come in on the evening service. How long has the company been operating that sector after dark?

Captain: The evening one started, what, last year some time. Middle of last year. Before that Tarlton was a middle-of-the-day run, in and out in daylight.

Second investigator: And did anything change for the crews when the evening sector came in? Any briefing, anything new?

Captain: Not that reached me. It just appeared on the roster. Same aeroplane, same aerodrome, later.

Lead investigator: I want to ask about the decision to stop. You've said you called V one and then stopped. Talk me through that.

Captain: Yeah. I know what the book says. Above V one you go. And I know that, I've known that for twenty years, and I stopped anyway. But what I had in front of me was two animals the size of a person on the runway, and I'm about to rotate into them at a hundred and twenty knots. If I take that on the gear at rotate, or through a prop, I'm airborne with a broken aeroplane over country with nowhere to go at night. So I took the one I could see the end of. I'm not going to sit here and tell you it was by the book. It wasn't.

Lead investigator: In your recurrent training, have you ever been given a stop or go scenario that wasn't an engine problem?

Captain: No. It's always an engine. Engine fire, engine failure, config warning, and they're all below V one anyway. That's the whole design of the exercise. Above V one in the sim you continue, because the thing they've given you is an engine, and the aeroplane flies on one. Nobody's ever put something on the runway.

Second investigator: The runway lighting that evening. How was it for you? Did it help or hinder?

Captain: It was bright. Honestly it was brighter than I'd want, and it's pilot-activated out there, so there's nobody to ask to turn it down. You've got this corridor of lights and everything outside it is black, so anything off the seal you don't see coming. I couldn't tell you if that's why I never picked them up. I was heads-in on the engine page after V one, which is where I'm meant to be.

Lead investigator: How were you feeling by that point in the day? Honestly.

Captain: Tired. It was eleven hours in and it was the fifth sector. But not - I wasn't struggling. The decision wasn't slow, if that's what you're asking. If you'd asked me to fly another sector I'd have said no. For the two seconds that mattered, I was there.

Lead investigator: Last one from me. Is there anything you think we should be looking at that we haven't asked about?

Captain: The list. Whatever the rule is that decides which aerodromes go on that list, go and look at it, because it's got Tarlton wrong and I doubt it's the only one.

Lead investigator: That's helpful. Thanks. I'll stop the recording there.

Interview transcript, first officer

The Teams transcript of the interview with the first officer, with every speaker identified by role.

Lead investigator: We're recording. You're happy with that, and you've got the support person outside if you want them in at any point?

First officer: Yes, that's fine.

Lead investigator: Same as we discussed. Tell me about the eighteenth in your own words, from wherever you'd like to start, and I'll save my questions.

First officer: OK. It was a long day but it was a normal day, right up until it wasn't. Five sectors, we picked up the last two. I'd only been into Tarlton twice before and both of those were daylight. So that was my first one out there in the dark. The captain flew us out there and I was flying the sector home. The brief was normal. We talked about the runway length because it's not long, and the figures, and the headwind, and that was it. Nothing about animals. I didn't think of it and he didn't raise it. I'd have followed his lead if he had. Power set, all normal. Eighty knots. And then somewhere just after V one I had two roos in the landing lights. They came in from the left. They were on the seal by the time I registered what I was looking at. I called them. I think I just said roos, roos. I don't remember what I said exactly. And he said stopping, I have control, and I gave him control, and then we hit one. It was loud. It was really loud, and there was a bang on the right that was different to the first one, and vibration. I was calling speeds. Sixty, fifty. And I could see the runway end lights and I remember saying end's coming up. We went off. It wasn't violent. It was like driving onto a dirt shoulder, just noisy. Then it was quiet and he did the shutdown and we got everyone off. That's what I've got.

Lead investigator: Thank you. Can I go back to the moment you saw them. How far out were they?

First officer: I've thought about this a lot. Two hundred metres, maybe. It could have been less. It was the length of a fair bit of runway, is the honest answer.

Lead investigator: And how clearly could you see them, once you had them?

First officer: Clearly. Completely clearly. Once they were on the seal they were lit up, I could see the two of them and I could see which way they were going. It was before that I couldn't see anything. Outside the lights it's just black. They didn't appear, they arrived, if that makes sense.

Second investigator: The captain told us he found the runway lighting quite bright that night. Was that your impression?

First officer: I wouldn't have said that. It looked normal to me. But I've got two landings there to compare it with and both of them were in daylight, so.

Second investigator: You're saying you're not well placed to judge whether it was unusual.

First officer: That's right. It didn't strike me as bright. It didn't strike me as anything. I was looking down the runway.

Lead investigator: Between your call and the captain taking control - are you able to put any number on how long that was?

First officer: No. It felt like nothing. It was, I called, he called, it was one thing, not two things.

Lead investigator: Had you and the captain briefed anything about who'd make a stop decision, or what you'd stop for?

First officer: We do the standard take-off brief, so it's the captain's decision to reject, below V one, and above V one we continue. That's the words. But we don't brief a list of things. It's just below V one, reject for the stated items. Nobody ever briefs what you do about something on the runway.

Second investigator: Is that something you'd been taught, in your training here or before?

First officer: In the sim it's always engines. Failure, fire, or a config warning on the roll. All of those come at you well before V one, every single time. I've never seen an obstruction scenario in a simulator anywhere. I don't think the box even has a way of putting something on the runway.

Lead investigator: Did you have any knowledge, from anywhere, that kangaroos were an issue at Tarlton Springs?

First officer: No, and that's the bit that's stayed with me. It turns out everybody knew. The ground crew out there told me afterwards it's every night, that's their words. And I've flown in there and I had no way of knowing that, because it's nowhere.

Lead investigator: Nowhere meaning - can you be specific about what you looked at before the flight?

First officer: The plan, the charts, the aerodrome directory entry, the company aerodrome brief page for Tarlton. None of them say anything about animals. The company page has got a note about the runway width and the turning bay. That's it.

Second investigator: I want to ask about after you stopped. Talk me through the disembarkation, anything you noticed.

First officer: It was calm. Genuinely calm. He made the announcement, the cabin crew got them up by rows, out the front door down the stairs. One thing. When I came back through the cabin the floor lights weren't all on. There's a strip down the aisle and there were sections of it out.

Second investigator: Were the emergency lights selected on at that point?

First officer: Yes, he'd armed them, and the ceiling ones were on. It was just the floor strip, there were gaps in it, two patches I think, towards the back. It didn't matter on the night because the cabin lights were still on and it was an orderly walk-off. But I noticed it. I raised it with the engineer the next morning. I don't know what came of that.

Lead investigator: How were you feeling at that point in the duty? It was sector five and you'd been on eleven hours.

First officer: Tired, but that's a normal five-sector day for me. I wasn't fighting it. I'd had a proper break in Port Carrick in the middle.

Lead investigator: Anything you'd want an investigation like this to look at?

First officer: Whether anybody at the company actually knew about the roos out there. Because the people on the ground knew, and it never got to us, and I'd like to know where it stopped.

Lead investigator: That's a fair question and we'll follow it. Thank you. Stopping the recording.